

Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
CSE 3102, Lab No. 07, Lab Exercise

Consider the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

1. Create a database called “CricketerStatsDB”. Write the corresponding query for this in your report.
2. Create a table titled “Players” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
player_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
player_name	VARCHAR (100)	NOT NULL
country	VARCHAR (50)	NOT NULL
date of birth	DATE	NOT NULL
batting style	VARCHAR (20)	NOT NULL
bowling style	VARCHAR (50)	NULLABLE

3. Create a table titled “Matches” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
match_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
match_date	DATE	NOT NULL
venue	VARCHAR (100)	NOT NULL
match_type	VARCHAR (20)	NOT NULL

4. Create a table titled “Performances” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
performance_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
player_id	INT	NOT NULL
match_id	INT	NOT NULL
runs_scored	INT	NOT NULL DEFAULT 0
balls_faced	INT	NOT NULL DEFAULT 0
wickets_taken	INT	NOT NULL DEFAULT 0
overs_bowled	DECIMAL (4,1)	NOT NULL DEFAULT 0.0

5. Populate the table “Players” with the following data. Write the corresponding insertion query for this in your report.

player_name	country	date of birth	batting style	bowling style
Virat Kohli	India	1988-11-05	Right-Handed	NULL
Joe Root	England	1990-12-30	Right-Handed	Off Spin
David Warner	Australia	1986-10-27	Left-Handed	NULL
Shaheen Afridi	Pakistan	2000-04-06	Left-Handed	Left-arm Fast
Kane Williamson	New Zealand	1990-08-08	Right-Handed	Off Spin

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6. Populate the table “Matches” with the following data. Write the corresponding insertion query for this in your report.

match_date	venue	match_type
2024-03-15	MCG, Australia	Test
2024-04-02	Lord’s, England	ODI
2024-04-18	Eden Gardens, India	T20
2024-05-01	Gaddafi Stadium, Pakistan	Test

7. Populate the table “Performances” with the following data. Write the corresponding insertion query for this in your report.

player_id	match_id	runs_scored	balls_faced	wickets_taken	overs_bowled
1	1	120	200	0	0.0
2	1	150	250	0	5.0
3	2	80	70	0	0.0
4	4	15	10	5	20.0
5	3	45	30	0	2.0
1	3	60	45	0	0.0
2	4	55	50	0	4.0

After creating and populating the tables, write queries in your report to do the following tasks.

- a) **Query 1:** Display all players.

Sample Output:

player_id	player_name	country	date_of_birth	batting_style	bowling_style
1	Virat Kohli	India	1988-11-05	Right-Handed	NULL
2	Joe Root	England	1990-12-30	Right-Handed	Off Spin
3	David Warner	Australia	1986-10-27	Left-Handed	NULL
4	Shaheen Afridi	Pakistan	2000-04-06	Left-Handed	Left-arm Fast
5	Kane Williamson	New Zealand	1990-08-08	Right-Handed	Off Spin

- b) **Query 2:** List all match venues and types.

Sample Output:

venue	match_type
MCG, Australia	Test
Lord’s, England	ODI
Eden Gardens, India	T20
Gaddafi Stadium, Pakistan	Test

- c) **Query 3:** Find players born after 1990.

Sample Output:

player_name	country	date_of_birth
Shaheen Afridi	Pakistan	2000-04-06

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d) Query 4: Show performances where runs scored > 50.

Sample Output:

player_id	match_id	runs_scored
1	1	120
2	1	150
1	3	60
2	4	55

e) Query 5: Find the number of matches played at each venue.

Sample Output:

venue	match_count
MCG, Australia	1
Lord's, England	1
Eden Gardens, India	1
Gaddafi Stadium, Pakistan	1

f) Query 6: List players with their total runs scored.

Sample Output:

player_name	total_runs
Virat Kohli	180
Joe Root	205
David Warner	80
Shaheen Afridi	15
Kane Williamson	45

g) Query 7: Find players who scored more than 100 in any match.

Sample Output:

player_name	runs_scored
Virat Kohli	120
Joe Root	150

h) Query 8: Find average balls faced per player.

Sample Output:

player_name	avg_balls
Virat Kohli	122.5
Joe Root	150
David Warner	70
Shaheen Afridi	10
Kane Williamson	30

i) Query 9: Find the top wicket-taker player.

Sample Output:

player_name	total_wickets
Shaheen Afridi	5

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j) Query 10: Find match-wise total runs scored and wickets taken by all players.

Sample Output:

match_id	venue	total_runs	total_wickets
1	MCG, Australia	270	0
2	Lord's, England	80	0
3	Eden Gardens, India	105	0
4	Gaddafi Stadium, Pakistan	70	5

Query Hints: You may use the following hints to solve the queries.

Query No	Concepts Used
1-2	Simple SELECT
3-4	SELECT + WHERE
5	GROUP BY
6-7	JOIN + GROUP/Condition
8	JOIN + AVG()
9	JOIN + Subquery + Aggregate
10	JOIN + Group multiple fields